



SHOMESWARAN A/L MUGUNTHAN

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SUMMARY

Software Development undergraduate with a CGPA of 3.91 and Dean's List achievements across all completed semesters. Experienced in developing full-stack web and mobile applications using Flutter, React, Firebase, Supabase, and REST APIs, with project experience in AI, machine learning, IoT, and database systems. Passionate about building scalable, user-focused software solutions and continuously expanding technical expertise.

EDUCATION

Universiti Teknikal Malaysia Melaka (UTeM) 2023 - Present

- Bachelor of Computer Science (Software Development) with Honours
- CGPA: 3.91

Kolej Matrikulasi Negeri Sembilan (KMNS) 2022 - 2023

- Physical Science Stream
- CGPA: 4.00 MUET: Band 4.0

Sekolah Menengah Kebangsaan Dato Mohd Said 2017 - 2022

- SIJIL PELAJARAN MALAYSIA (SPM)
- Result: 8A 2B

SKILLS

- Programming Languages:** Java, Python, JavaScript, TypeScript, Dart, PHP, SQL, C++, HTML, CSS
- Frameworks & Technologies:** Flutter, React.js, React Native, Node.js, FastAPI, REST APIs
- Databases & Cloud:** Firebase, Supabase, MySQL, PostgreSQL
- AI & IoT:** Machine Learning, NumPy, Pandas, Arduino, ESP32
- Tools & Platforms:** Git, GitHub, Visual Studio Code, Android Studio, Figma, Canva, XAMPP
- Development Skills:** Full-Stack Web Development, Mobile Application Development, API Development, Database Design, UI/UX Design, Testing, Debugging, System Design, Technical Documentation

LANGUAGE PROFICIENCIES

- Bahasa Melayu (Fluent)
- English (Fluent)
- Tamil (Native)

PROJECTS

NutriPulse AI — AI-Powered Health and Nutrition Application

Flutter | Firebase | Machine Learning | IoT Health Integration (ESP32)

- Developed a full-stack mobile healthcare application integrating AI-assisted nutrition tracking, food recognition, and IoT health monitoring.
- Built secure Firebase Authentication, Firestore database integration, and real-time health record management.
- Implemented machine learning-based meal recognition and personalised nutrition recommendations using health and lifestyle data.
- Developed personalised recommendations based on users' nutrition records, health data, and lifestyle activities.

Farm2Screen — Pineapple Marketplace and Information Application (SULAM)

Flutter | Firebase | Supabase Storage | Mobile Application Development

- Developed a mobile marketplace application connecting pineapple entrepreneurs with customers through dedicated entrepreneur and customer portals.
- Implemented product management, customer ordering, shopping cart, checkout, order history, and order-status tracking features.
- Integrated Firebase Authentication and Firestore for secure user access, real-time data management, and role-based functionality.
- Used Supabase Storage for product image uploads and developed sales analytics, announcements, and daily sales summary features for entrepreneurs.

IPETRO VIRS — Visual Inspection Reporting System

JavaScript | HTML | CSS | SQL |

- Web Application Developed a web-based inspection reporting system based on API 510 inspection requirements.
- Implemented inspection data management and automated report generation with PDF and Microsoft Word export.
- Built separate frontend and backend modules to support structured inspection workflows and database operations.
- Collaborated with a six-member development team and deployed the completed system as a live web application.

Retinal Disease Detection System

Python | Machine Learning | Image Processing

- Developed an image-based machine learning system to support early retinal disease screening.
- Applied image preprocessing and classification techniques to analyse retinal fundus images.
- Implemented a prediction workflow that assists users in identifying possible retinal abnormalities.
- Designed the system to support healthcare-related decision-making through automated medical image analysis.

JPJ Management System

Console Application | File Handling | Vehicle Management

- Developed a console-based vehicle management system supporting vehicle record administration.
- Implemented insert, delete, update, search, and sorting functions for vehicle information.
- Added vehicle maintenance record management and PUSPAKOM inspection-check features.
- Designed a structured menu-based interface to improve data organisation and usability.

EXTRACURRICULAR ACTIVITIES

• Mandarin Language Video Project — Director & Scriptwriter

Directed the production of a Mandarin-language video and developed the script, while coordinating team members, filming, presentation, pronunciation, and video-editing activities.

• Softball — Co-curricular Participant

Participated in softball training and a small tournament, developing teamwork, discipline, communication, and sportsmanship skills.

CERTIFICATIONS & RECOGNITION

- Dean's List Award – Achieved the Dean's List for all completed semesters.
- HackerRank Certifications – Software Engineer and Python (Basic), covering problem-solving, SQL, REST APIs and Python fundamentals.
- Cisco Certifications – C++ Advanced and Python Essentials 1.
- IBM Certification – Code Generation and Optimization Using IBM Granite.
- Oracle Database Foundations – Completed through Oracle Academy.
- Linear Algebra with MATLAB – Course completion certificate from MathWorks.
- Matriculation Academic Excellence – Graduated with a CGPA of 4.00 and MUET Band 4.0.
- Anugerah Murid Cemerlang – Recognised for achieving 8As in SPM 2021.